

Total No. of Questions : 8]

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Roll No

AU/ME-221-CBCS

B.E., III Semester

Examination, June 2020

Choice Based Credit System (CBCS)

Strength of Materials

Time : Three Hours

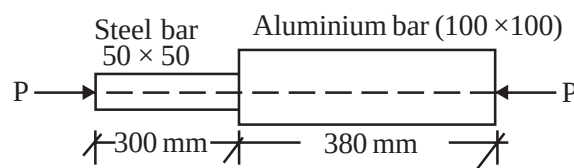
Maximum Marks : 60

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

iii) Draw neat and clean sketches. Assume data suitably.

1. a) Derive from fundamental, the relation for the deformation of a body, when it is subjected to
 - i) tensile force and
 - ii) its own weight
- b) A member formed by connecting a steel bar to an aluminium bar as shown in fig.



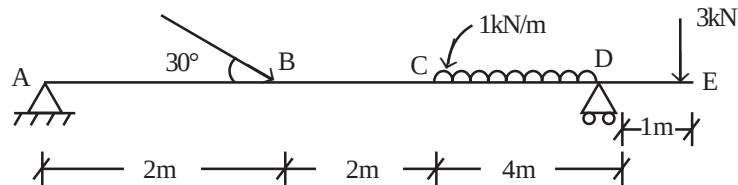
Assuming that the bars are prevented from buckling sidewise, calculate the magnitude of force P , that will cause the total length of the member to decrease by 0.25mm. $E_S = 210 \text{ GPa}$ $E_A = 70 \text{ GPa}$

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2. a) Analyse the beam shown in fig and draw shear force diagram.



- b) Illustrate the term 'beam of uniform strength'. Explain its necessity.
3. a) Prove $\frac{Z}{R} = \frac{c \theta}{l}$ in case of torsion of a circular shaft.
- b) Write the assumptions for finding out the shear stress in circular shaft, subjected to torsion.
4. a) A machine member is subjected to the following stresses $\sigma_x = 150$ MPa, $\tau_{xy} = 24$ MPa. Find the equivalent stress as per the following theories of failure:
- Shear stress theory
 - Normal stress theory
- b) Explain any four theories of failure.
5. a) What do you understand by the term 'column' and 'strut'? Distinguish clearly between long columns and short columns.
- b) Describe the assumptions in the Euler's column theory.
6. a) Draw the Shear Force and Bending Moment diagram for a cantilever of length L carrying a point load W at the free end.
- b) State Maxwell's reciprocal theorem.

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7. a) Calculate:
- i) the change in diameter
 - ii) the change in length of a thin cylindrical shell 100cm diameter 1cm thick and 5m long when subjected to internal pressure of 3N/mm^2 .
Take $E = 2 \times 10^5 \text{ N/mm}^2$, Poisson's Ratio = 0.3.
- b) How will you use conjugate beam method for finding slope and deflection at any section of given beam.
8. Write short notes on:
- a) Flitched Beam
 - b) Principal Planes
 - c) Assumptions made in Lamé's theory
 - d) Thin Cylinders
